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Integrating Ticketing Systems with Security Alerting for Automated Case Management

ABSTRACT

Modern organisations depend on computer systems and networks to carry out their daily activities. As cyberattacks continue to increase, it has become important to monitor systems continuously so that suspicious activities can be detected before they cause damage. One common attack against Linux servers is the Secure Shell (SSH) brute-force attack, where an attacker repeatedly tries different username and password combinations until the correct password is found. Detecting these attacks manually is difficult because operating systems generate thousands of log entries every day.

This project presents the design and implementation of a small virtual Security Operations Centre (SOC) using VMware Workstation, Kali Linux, Ubuntu Server and Graylog SIEM. Kali Linux was used to simulate an attacker by performing SSH brute-force attacks using Hydra. Ubuntu Server acted as the target system and generated authentication logs whenever login attempts were made. These logs were forwarded to Graylog using rsyslog, allowing Graylog to analyse them in real time.

An Event Definition was created in Graylog to detect repeated failed SSH login attempts within a specified time window. When the configured threshold was exceeded, Graylog automatically generated a security event and an alert. This demonstrated how automated monitoring can reduce the time required to identify suspicious activities.

The project also explored integrating Graylog with TheHive for automated incident response. Although the complete deployment of TheHive could not be achieved because of hardware limitations, the proposed architecture demonstrates how security alerts can automatically create investigation cases in a production environment.

Overall, the project successfully demonstrated that centralized log management and automated security alerting improve the detection of cyberattacks and provide a strong foundation for effective incident response.

Keywords: Graylog, SIEM, Cybersecurity, SSH, Hydra, rsyslog, Incident Response, Log Management.

Contents

ABSTRACT	2
CHAPTER ONE.....	5
INTRODUCTION.....	5
1.1 Background of the Study	5
1.2 Problem Statement	6
1.3 Aim of the Project.....	6
1.4 Objectives of the Project	6
1.5 Significance of the Study	6
CHAPTER TWO.....	7
LITERATURE REVIEW AND TOOLING REVIEW	7
2.1 Introduction	7
2.2 Virtualisation	7
2.3 Ubuntu Server	8
2.4 Kali Linux.....	8
2.5 Secure Shell (SSH)	8
2.6 Hydra	9
2.7 rsyslog	9
2.8 Graylog.....	9
2.10 Alerts	10
2.11 TheHive	11
2.12 Relationship Between the Tools.....	11
2.13 Chapter Summary	12
CHAPTER THREE	12
METHODOLOGY.....	12
3.1 Introduction	12
3.2 Research Method.....	12
3.3 Development Environment	13
3.4 Virtual Laboratory Design	13

3.5 Hardware Requirements	14
3.6 Software Requirements	14
3.7 Network Configuration	15
3.8 System Architecture.....	15
3.9 Network Configuration	17
3.9 Graylog Installation	17
3.10 Chapter Summary	23
CHAPTER FOUR	23
SYSTEM IMPLEMENTATION, RESULTS AND DISCUSSION.....	23
4.1 Introduction	23
4.2 Virtual Laboratory Setup	24
4.3 Planned Integration with TheHive	24
4.4 Challenges Encountered	25
4.5 Lessons Learned	26
4.6 Chapter Summary	27
CHAPTER FIVE	27
CONCLUSION AND RECOMMENDATIONS	27
5.1 Introduction	27
5.2 Discussion	27
5.3 Achievement of Objectives.....	28
5.4 Challenges Encountered	28
5.5 Lessons Learned	29
5.6 Recommendations.....	30
5.7 Future Work	30
5.8 Conclusion	30
REFERENCES.....	31
APPENDICES.....	31

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Cybersecurity has become one of the biggest concerns for organisations because computers and networks are constantly exposed to different types of cyberattacks. Businesses, universities, banks, hospitals, and government institutions all rely on computer systems to store and process important information. As a result, attackers continuously look for weaknesses that they can exploit to gain unauthorised access, steal sensitive information, or disrupt normal operations.

One common attack used by hackers is the **SSH brute-force attack**. In this type of attack, an attacker repeatedly tries different username and password combinations until the correct login credentials are found. If the attack is successful, the attacker gains access to the target system and may perform malicious activities such as stealing files, installing malware, or creating new user accounts.

Because thousands of events are generated on computer systems every day, it is difficult for security administrators to manually examine every log file. This challenge has led to the development of **Security Information and Event Management (SIEM)** systems. SIEM solutions collect logs from different devices, analyse them, identify suspicious activities, and generate alerts whenever potential security incidents are detected.

Graylog is one of the SIEM platforms widely used for centralised log management and security monitoring. It allows security analysts to collect logs from multiple sources, create detection rules, monitor events in real time, and generate alerts whenever suspicious behaviour is identified. These features help organisations detect cyberattacks much faster than relying on manual log analysis.

Although detecting attacks is important, responding to them is equally necessary. Modern Security Operations Centres (SOCs) often integrate SIEM platforms with incident response or case management systems. One example of such a system is **TheHive**, which allows security teams to create investigation cases, assign tasks to analysts, document findings, and track incident response activities from beginning to end.

This project focuses on designing and implementing a virtual cybersecurity laboratory that demonstrates how a SIEM platform can detect SSH brute-force attacks and generate security alerts. The project also presents the planned integration between Graylog and TheHive to automate incident response. Due to the limited hardware resources available during implementation, the complete deployment of TheHive could not be completed locally. However, the project successfully demonstrates the attack detection and alerting workflow while presenting the case management integration as the intended extension of the system.

1.2 Problem Statement

Many organisations still rely on manually reviewing system logs to identify suspicious activities. As the number of servers and users increases, the volume of generated logs also increases. This makes manual monitoring inefficient because security analysts may overlook important warning signs.

An SSH brute-force attack produces many failed login attempts within a short period. If these failed attempts are not detected early, an attacker may eventually gain access to the server. Detecting such attacks manually requires searching through large log files, which is both slow and error-prone.

This project addresses this problem by implementing a system that automatically collects logs, analyses them, detects suspicious login attempts, and generates alerts without requiring manual monitoring.

1.3 Aim of the Project

This project aims to design and implement a security monitoring system that automatically detects SSH brute-force attacks using Graylog SIEM and demonstrates how the generated alerts can support automated incident response.

1.4 Objectives of the Project

The objectives of this project are to:

- Design a virtual cybersecurity laboratory.
- Configure Ubuntu Server as the target machine.
- Configure Kali Linux as the attacker machine.
- Install and configure Graylog.
- Configure rsyslog to forward authentication logs.
- Simulate an SSH brute-force attack using Hydra.
- Configure Graylog to detect repeated failed login attempts.
- Generate automated security alerts.
- Demonstrate the proposed integration between Graylog and TheHive.

1.5 Significance of the Study

This project focuses on monitoring SSH authentication logs generated by an Ubuntu Server. The implementation is limited to detecting brute-force attacks performed using Hydra within a controlled virtual laboratory. The project demonstrates attack detection and alert generation using Graylog SIEM. TheHive integration is presented as a proposed extension because the available hardware could not support its full deployment.

1.6 Report Organisation

This report is organised into seven chapters.

Chapter One introduces the project, including the background, problem statement, objectives, scope, and significance.

Chapter Two reviews the tools and technologies used in the project.

Chapter Three describes the methodology and the design of the virtual laboratory.

Chapter Four explains the implementation process.

Chapter Five presents the results obtained during testing.

Chapter Six analyses the results and provides recommendations.

Chapter Seven concludes the project and suggests areas for future work.

CHAPTER TWO

LITERATURE REVIEW AND TOOLING REVIEW

2.1 Introduction

Before building any cybersecurity monitoring system, it is important to understand the technologies that make the system work. This project combines several open-source tools to create a small Security Operations Centre (SOC) capable of detecting SSH brute-force attacks. Each tool performs a specific task, and together they create a complete monitoring and alerting workflow.

This chapter reviews the main technologies used during the implementation and explains their role within the project

2.2 Virtualisation

Virtualisation is the process of creating virtual computers inside a physical computer. Instead of buying several computers, one computer can run multiple virtual machines, each with its own operating system.

In this project, **VMware Workstation** was used to create the virtual laboratory. This allowed Kali Linux, Ubuntu Server and the planned TheHive server to run independently while sharing the same physical computer.

Using virtual machines also made it possible to safely simulate cyberattacks without affecting the host computer or any real network.

Role in this project

VMware Workstation provided the environment where the entire project was developed. It allowed communication between the attacker machine and the target server while keeping the laboratory isolated from external systems.

2.3 Ubuntu Server

Ubuntu Server is a Linux operating system designed for hosting applications and network services. It is widely used because it is stable, secure and open source.

Unlike the desktop version of Ubuntu, Ubuntu Server does not focus on graphical applications. Instead, it is managed mainly through the command line.

In this project, Ubuntu Server performed two important roles.

First, it acted as the **target machine** that received SSH login requests from Kali Linux.

Secondly, it hosted the **Graylog SIEM**, which monitored the authentication logs generated by the operating system.

Whenever someone attempted to log in through SSH, Ubuntu automatically recorded the activity in its authentication log file.

Those logs became the source of information analysed by Graylog

2.4 Kali Linux

Kali Linux is a Linux distribution developed specifically for cybersecurity testing and penetration testing. It contains hundreds of security tools used to assess the security of computer systems.

In this project, Kali Linux was **not used to attack a real organisation**. Instead, it was used to simulate an attacker within a controlled laboratory environment.

The purpose was to generate realistic attack data that could test whether the monitoring system was functioning correctly.

Among the many tools available in Kali Linux, Hydra was selected to perform the SSH brute-force simulation.

2.5 Secure Shell (SSH)

Secure Shell (SSH) is a protocol that allows users to connect securely to remote computers over a network. System administrators commonly use SSH to manage Linux servers without being physically present.

Every time a user attempts to log in through SSH, Ubuntu records the activity in its authentication log. These logs contain information such as:

- Username
- Source IP address
- Login time
- Whether the login was successful or failed

Because SSH generates detailed authentication records, it is an excellent source of information for detecting brute-force attacks.

In this project, SSH provided the service that Hydra attempted to access during the attack simulation

2.6 Hydra

Hydra is an open-source password testing tool that can perform login attempts against many different network services.

In this project, Hydra was used to simulate an attacker by repeatedly trying different username and password combinations against the Ubuntu Server's SSH service.

The purpose was **not to gain access**, but to generate multiple failed login attempts that could test the monitoring system.

Each failed login attempt produced an authentication log, allowing Graylog to analyse the activity and determine whether it matched the configured detection rules.

Without Hydra, it would have been difficult to generate enough authentication events to properly test the monitoring system

2.7 rsyslog

Operating systems continuously generate logs whenever activities occur.

However, simply storing logs on one server does not provide centralized monitoring.

This is where **rsyslog** becomes important.

rsyslog is a log forwarding service that collects system logs and sends them to another server.

Think of rsyslog as a delivery service.

Whenever Ubuntu created a new authentication log, rsyslog automatically forwarded a copy of that log to Graylog.

Without rsyslog, Graylog would never receive the authentication logs generated by Ubuntu.

Its role in this project was therefore to connect the target server with the monitoring server.

2.8 Graylog

Graylog is an open-source Security Information and Event Management (SIEM) platform used for collecting, storing, searching and analysing log data.

In this project, Graylog acted as the **security monitoring system**.

Instead of reading thousands of authentication logs manually, Graylog automatically received the logs forwarded by rsyslog and analysed them continuously.

One of Graylog's most important features is the ability to create **Event Definitions**.

An Event Definition is simply a rule.

For example, during this project the following rule was created:

If more than five failed SSH login attempts occur within one minute, create an event.

Whenever this condition became true, Graylog generated an Event.

The Event then triggered an Alert, notifying the administrator that suspicious login activity had been detected.

Graylog therefore became the central component of the entire monitoring system.

2.9 Event Definitions

Event Definitions are one of the most powerful features available in Graylog.

They allow administrators to define the conditions that indicate suspicious behaviour.

Rather than monitoring every log individually, Graylog continuously compares incoming logs against these predefined rules.

For this project, the Event Definition counted failed SSH login attempts within a one-minute period.

If the number of failures exceeded the configured threshold, Graylog created an Event.

This approach reduced false alarms while ensuring genuine brute-force attacks were detected quickly.

2.10 Alerts

An Alert is a notification generated after an Event has been detected.

The purpose of an alert is to immediately inform administrators that something important has happened.

Without alerts, administrators would still need to regularly check Graylog dashboards to discover suspicious activities.

In this project, Graylog automatically generated an alert whenever the Event Definition detected repeated failed SSH login attempts.

This demonstrated how automated monitoring reduces the time required to identify potential attacks.

2.11 TheHive

TheHive is an open-source Security Incident Response Platform (SIRP) used by Security Operations Centres (SOCs) to manage investigations.

Unlike Graylog, which focuses on detecting attacks, TheHive focuses on managing what happens **after** an attack has been detected.

Its main features include:

- Creating investigation cases
- Assigning incidents to analysts
- Tracking investigation progress
- Recording evidence
- Producing investigation reports

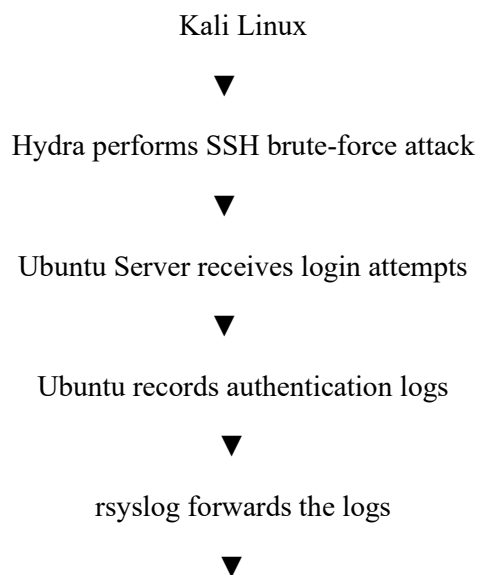
The original objective of this project was to integrate Graylog with TheHive so that every important alert would automatically create an investigation case.

Although the deployment could not be completed because of hardware limitations, the proposed integration demonstrates how the monitoring system could be expanded into a complete incident response platform.

2.12 Relationship Between the Tools

Although each tool performs a different task, they work together as one complete monitoring system.

The workflow is shown below.



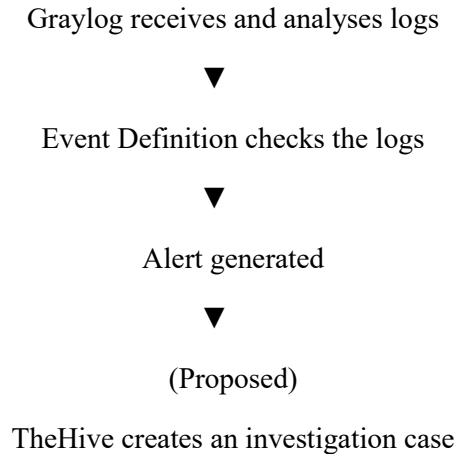


Figure 2.1: Overall Workflow of the Monitoring System

2.13 Chapter Summary

This chapter reviewed the technologies used in the implementation of the project. VMware Workstation provided the virtual environment, Ubuntu Server generated authentication logs, Kali Linux simulated the attacker, Hydra performed the SSH brute-force attack, rsyslog forwarded system logs, and Graylog analysed the logs to detect suspicious activities and generate alerts. TheHive was introduced as the planned incident response platform that would manage security investigations after alerts were generated.

Together, these tools formed an integrated security monitoring workflow capable of detecting SSH brute-force attacks in a controlled virtual environment.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methods used to design and implement the cybersecurity monitoring system. It explains how the virtual laboratory was created, the hardware and software used, the network configuration, and the steps followed to build the final monitoring solution.

The implementation was carried out in a controlled virtual environment to ensure that all attack simulations could be performed safely without affecting real systems.

3.2 Research Method

This project used an **experimental research approach**.

An experimental approach was chosen because it allowed a complete cybersecurity monitoring system to be built and tested in a controlled environment. Instead of studying existing attacks, the project generated its own attack traffic using Kali Linux and Hydra. This made it possible to observe how the monitoring system responded to real authentication events.

The implementation followed these main stages:

1. Build the virtual laboratory.
2. Configure the Ubuntu Server.
3. Install and configure Graylog.
4. Configure log forwarding using rsyslog.
5. Simulate an SSH brute-force attack.
6. Analyse logs in Graylog.
7. Generate automated alerts.

3.3 Development Environment

The project was developed using VMware Workstation running on a Windows computer.

The host computer had **8 GB of RAM**, so careful allocation of resources was necessary to ensure that the virtual machines could run without affecting the performance of the host operating system.

The virtual environment allowed different machines to communicate as though they were connected to the same local network.

3.4 Virtual Laboratory Design

Three virtual machines were prepared during the project.

Virtual Machine 1 – Kali Linux

Kali Linux was configured as the attacker machine.

Its main purpose was to simulate a hacker attempting to gain access to the Ubuntu Server through SSH.

The Hydra tool installed on Kali Linux generated repeated login attempts using different password combinations.

This machine produced the attack traffic that tested the monitoring system.

Virtual Machine 2 – Ubuntu Server

Ubuntu Server acted as the target machine.

The OpenSSH service was installed to allow remote logins.

Whenever someone attempted to log in through SSH, Ubuntu automatically recorded the activity in its authentication logs.

The same Ubuntu Server also hosted the Graylog SIEM platform.

Using a single server for both purposes reduced the amount of memory required while still allowing the monitoring system to function correctly.

Virtual Machine 3 – Planned TheHive Server

A third virtual machine was prepared for deploying TheHive.

The intention was to integrate Graylog with TheHive so that every security alert generated by Graylog would automatically create an investigation case.

During implementation, deployment was attempted using Docker.

However, the supporting services required by TheHive consumed more memory than was available on the host computer.

As a result, Cassandra and Elasticsearch failed to start correctly, preventing the complete deployment of TheHive.

Although the integration could not be completed, the planned architecture is included as part of the proposed solution.

3.5 Hardware Requirements

The implementation was carried out on the following hardware.

Component	Specification
Host Computer	Windows PC
Processor	Intel Processor
Memory	8 GB RAM
Storage	SSD
Virtualisation Software	VMware Workstation

3.6 Software Requirements

Software	Purpose
VMware Workstation	Create virtual machines

Ubuntu Server 24.04	Target server
Kali Linux	Attack simulation
Graylog	Security monitoring
rsyslog	Log forwarding
Hydra	SSH brute-force simulation
Docker	Deploy Graylog services
OpenSSH Server	Remote login service

3.7 Network Configuration

The virtual machines were connected using both **NAT** and **Host-only** networking.

NAT Network

The NAT adapter provided Internet access.

This allowed software packages such as Graylog, Docker and Ubuntu updates to be downloaded during installation.

Host-only Network

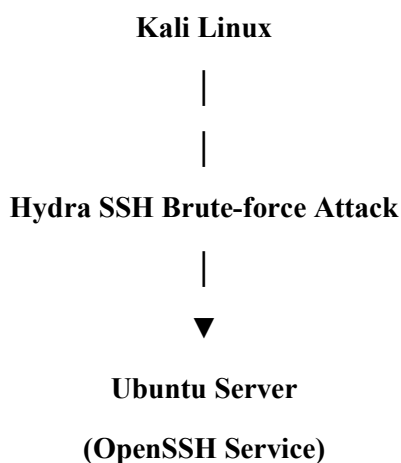
The Host-only adapter allowed communication between the virtual machines without exposing the laboratory to external networks.

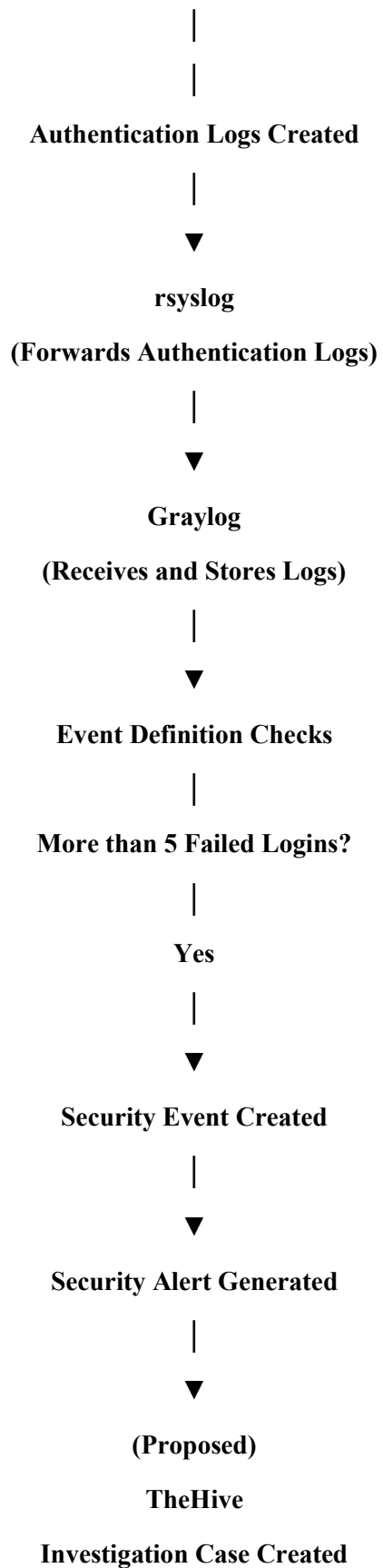
This ensured that the attack simulation remained completely isolated.

3.8 System Architecture

The monitoring system was designed using a simple client–server architecture.

The complete workflow is shown below.





3.9 Network Configuration

The virtual laboratory used both NAT and Host-only networking.

NAT networking allowed Internet connectivity for software installation.

Host-only networking enabled secure communication between the virtual machines.

Example IP addressing:

Device	Example Address
Kali Linux	192.168.162.130
Ubuntu Server	192.168.101.137

The exact addresses may vary depending on the VMware network configuration.

3.9 Graylog Installation

Graylog was installed using Docker.

The installation included:

- MongoDB
- OpenSearch/Data Node
- Graylog

After installation:

The Graylog web interface became accessible through a web browser.

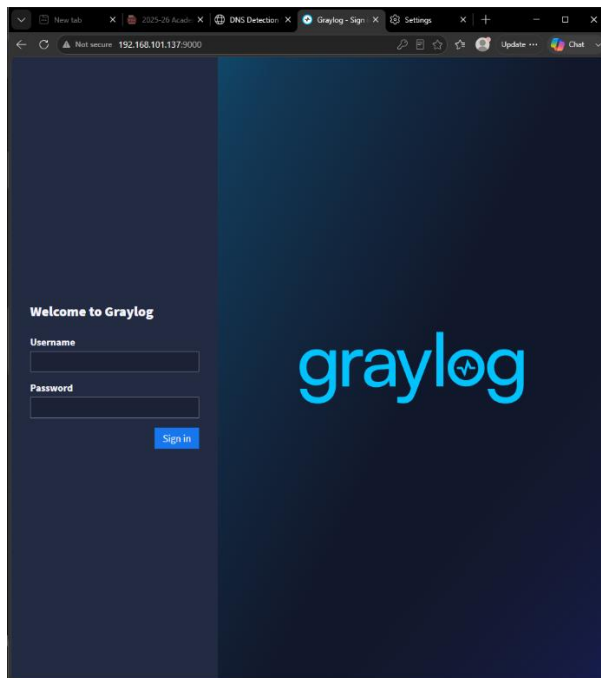


Figure 3.2

Graylog Login Page

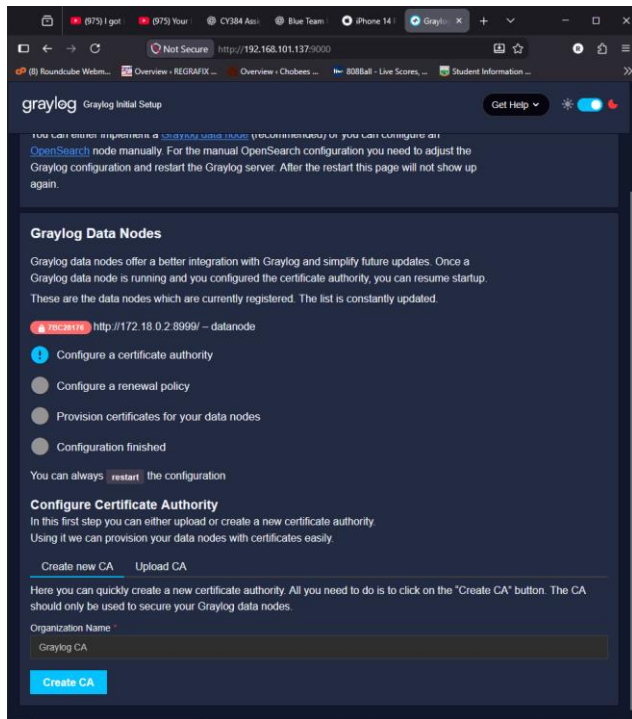
3.10 Configuring rsyslog

The Ubuntu Server was configured to forward authentication logs to Graylog using rsyslog.

The rsyslog configuration file was modified to send SSH authentication logs to the Graylog Syslog Input.

Figure 3.3

rsyslog Configuration



3.11 Graylog Inputs

A Syslog Input was created inside Graylog.

Configuration included:

Protocol

- UDP

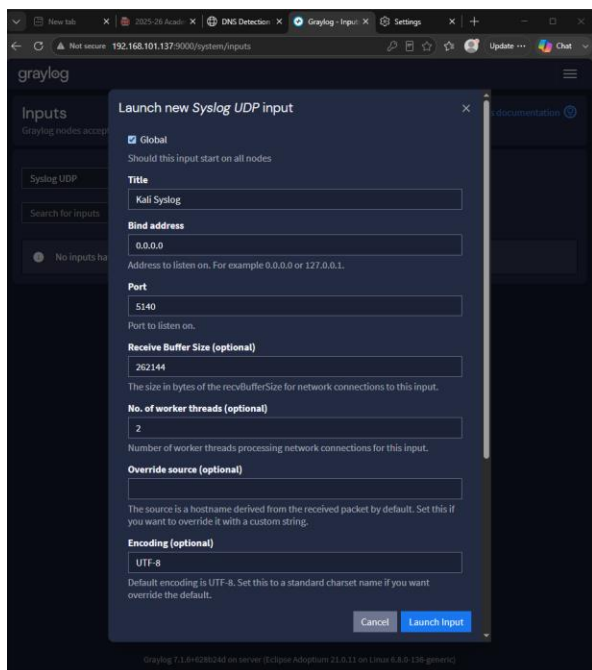
Port

- 5140

After activation, Graylog successfully received logs from the Ubuntu Server.

Figure 3.4

Graylog Syslog Input



3.12 Log Verification

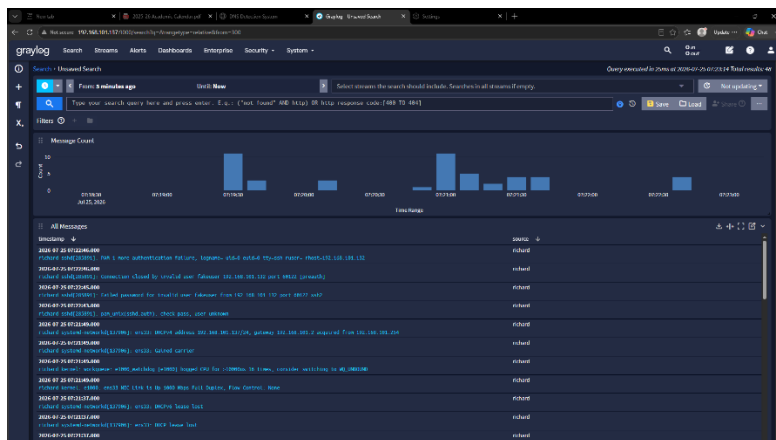
Once log forwarding was configured successfully, authentication events began appearing inside Graylog.

The received logs included:

- Failed login attempts
- Successful logins
- SSH authentication messages
- System events

Figure 3.5

Graylog Search Results



3.13 Event Definitions

An Event Definition was created to detect SSH brute-force attacks.

Configuration:

Search Filter

Authentication logs

Aggregation

Count messages

Threshold

More than five failed login attempts

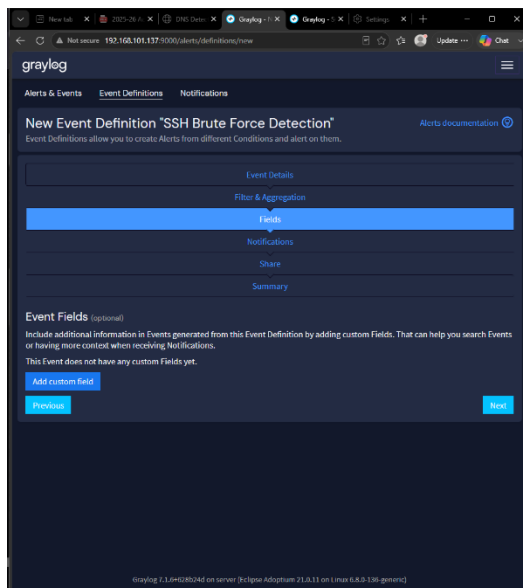
Time Window

One minute

Whenever this condition was satisfied, Graylog automatically generated a security event.

Figure 3.6

Graylog Event Definition



3.14 Alert Configuration

The Event Definition was connected to an Alert.

Whenever suspicious login activity exceeded the configured threshold, Graylog automatically generated an alert.

This reduced the need for manual log monitoring.

Figure 3.7

Graylog Alert Configuration

Event Condition
Define how Events of this type should be created. You can later use those Events as input on other Conditions, making it possible to build powerful Conditions based on others.

Condition Type
Filter & Aggregation
Choose the type of Condition for this Event.

Filter
Add information to filter the log messages that are relevant for this Event Definition.

Search Query
message:"Failed password"
Search query that Messages should match. You can use the same syntax as in the Search page, including declaring Query Parameters from Lookup Tables by using the `$parameterName` syntax.

Streams (Optional)
Select streams the search should include. Searches in all streams if empty.

Search within the last
5 minutes

☐ Use Cron Scheduling
Schedule this event with a Quartz cron expression.

Execute search every
5 minutes

☒ Enable
Should this event definition be executed automatically?

Create Events for Definition If...
☒ Filter has results
☐ Aggregation of results reaches a threshold

Event Limit
100
Maximum number of events to be created per execution of this event definition. Excess events will be suppressed.

Filter Preview

Timestamp	Message
2026-07-25T08:00:07.000Z	richard.suhl@2938181: Failed password for invalid user fakeuser from 192.168.101.132 port 32970 ssh2
2026-07-25T07:59:58.000Z	richard.suhl@2938181: Failed password for invalid user fakeuser from 192.168.101.132 port 32970 ssh2
2026-07-25T08:00:02.000Z	richard.suhl@2938181: Failed password for invalid user fakeuser from 192.168.101.132 port 32970 ssh2
2026-07-24T08:00:17.000Z	richard.suhl@2938181: Failed

3.15 SSH Brute-force Simulation

The attack was performed from the Kali Linux virtual machine using Hydra.

Hydra repeatedly attempted multiple username and password combinations against the Ubuntu SSH server.

This generated numerous authentication failures, which were forwarded to Graylog.

Figure 3.8

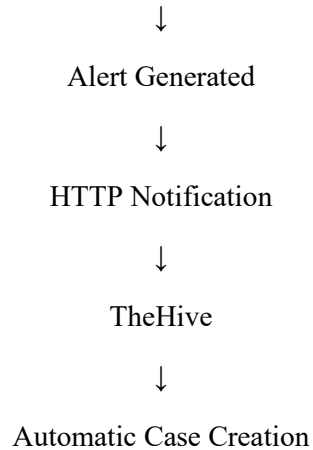
Hydra Attack

3.16 Planned TheHive Integration

The original design included integrating Graylog with TheHive through HTTP Event Notifications.

The intended workflow was:

Graylog



Deployment of TheHive was attempted during implementation. However, the production deployment required Cassandra and Elasticsearch services, which exceeded the available hardware resources of the testing environment.

Despite this limitation, the architecture demonstrates how automated case management would be integrated in a production environment.

3.10 Chapter Summary

This chapter described the design and implementation of the virtual cybersecurity monitoring laboratory. It presented the hardware and software requirements, virtual machine configuration, network architecture, Graylog deployment, log forwarding process, attack simulation, and event detection workflow. The implementation successfully demonstrated centralized log collection and automated security alert generation, providing the foundation for evaluating the system in the next chapter.

CHAPTER FOUR

SYSTEM IMPLEMENTATION, RESULTS AND DISCUSSION

4.1 Introduction

This chapter explains how the cybersecurity monitoring system was implemented and evaluates the results obtained during testing. The implementation involved configuring a virtual laboratory, installing Graylog, configuring log forwarding, simulating SSH brute-force attacks, and creating automated detection and alerting rules.

The purpose of the implementation was to determine whether Graylog could successfully detect SSH brute-force attacks and generate alerts that could support an incident response process.

4.2 Virtual Laboratory Setup

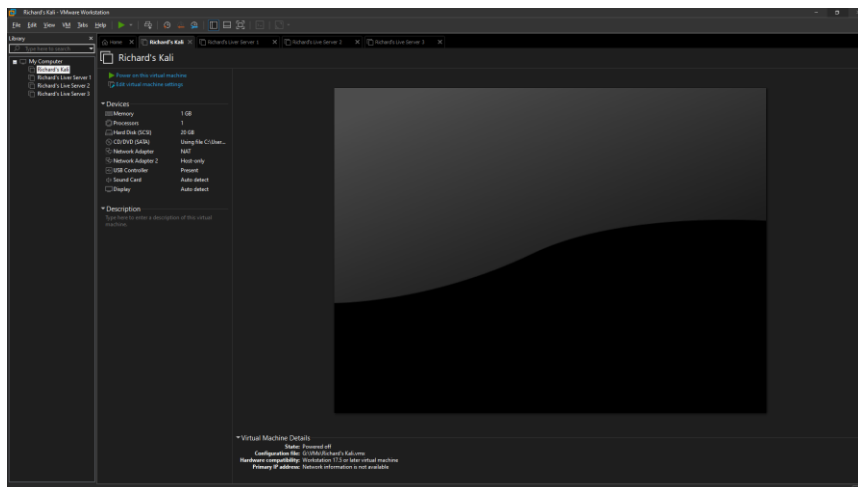
The project was implemented using VMware Workstation. Three virtual machines were created to simulate a small enterprise network.

The laboratory consisted of:

Virtual Machine	Function
Kali Linux	Attacker Machine
Ubuntu Server	Target Server and Graylog SIEM
Ubuntu Server	Planned TheHive Server

The Ubuntu Server hosting Graylog also acted as the SSH server that received the attacks generated from Kali Linux.

Figure 4.1: VMware Virtual Machines



4.3 Planned Integration with TheHive

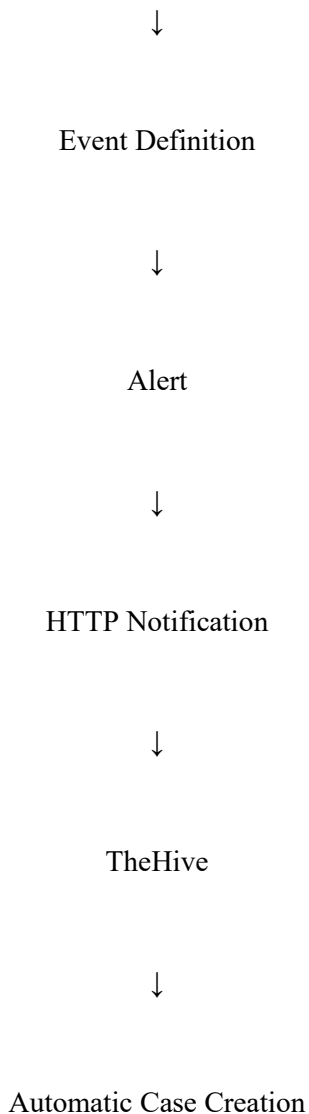
The original objective included integrating Graylog with TheHive to automate case creation after detecting security incidents.

The planned workflow was:

SSH Attack



Graylog Detection



During implementation, TheHive deployment was attempted using the official Docker-based installation.

However, the deployment could not be completed because the supporting services (Cassandra and Elasticsearch/OpenSearch) required more memory than was available in the laboratory environment.

The operating system repeatedly terminated these services because of insufficient memory, preventing TheHive from becoming operational.

Although the complete integration could not be demonstrated, the system architecture and implementation process clearly show how Graylog alerts would be forwarded to TheHive in a production environment.

4.4 Challenges Encountered

Several challenges were encountered during implementation.

Limited Hardware Resources

The host computer had 8 GB of RAM. Running multiple virtual machines together with memory-intensive services such as Cassandra and Elasticsearch exceeded the available system memory.

Docker Deployment Issues

Several Docker containers failed to start because backend services became unhealthy after running out of memory.

Elasticsearch and Cassandra Failures

The Linux kernel repeatedly terminated Java processes associated with Cassandra and Elasticsearch due to insufficient memory.

These failures prevented the complete deployment of TheHive.

Network Configuration

Initially, communication between the virtual machines was unsuccessful because of incorrect VMware network settings.

After configuring NAT and Host-only adapters correctly, communication between the virtual machines was restored.

Graylog Configuration

The Event Definition initially failed because the aggregation condition was not configured correctly.

After modifying the aggregation settings and threshold conditions, Graylog successfully generated events and alerts.

4.5 Lessons Learned

The implementation provided practical experience in several cybersecurity areas, including:

- Linux system administration
- SIEM deployment
- Centralized log management
- SSH security monitoring
- Attack simulation

- Event correlation
- Security alert generation
- Virtual network configuration

The project also demonstrated the importance of hardware planning when deploying enterprise security platforms.

4.6 Chapter Summary

This chapter presented the implementation of the cybersecurity monitoring system and discussed the results obtained during testing. The system successfully collected authentication logs, detected SSH brute-force attacks, and generated automated security alerts using Graylog.

Although the planned integration with TheHive could not be completed due to hardware limitations, the implementation achieved the primary objective of demonstrating centralised log management, attack detection, and automated security alerting.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter summarises the findings of the project, discusses whether the objectives were achieved, highlights the challenges encountered during implementation, and provides recommendations for future improvements.

5.2 Discussion

This project aimed to design and implement a virtual cybersecurity monitoring environment capable of detecting SSH brute-force attacks using Graylog and demonstrating how security alerting can support automated incident response.

The implementation was carried out successfully using a virtual laboratory environment consisting of Kali Linux, Ubuntu Server, and Graylog. The Ubuntu Server functioned as both the target system and the centralised log monitoring server. Kali Linux was used to simulate attacks using Hydra, while Graylog collected, analysed, and generated alerts based on authentication logs received from the target system.

The experimental results showed that Graylog was able to receive authentication logs forwarded through rsyslog in real time. The Event Definition configured in Graylog successfully detected repeated failed login attempts, while the Alert system automatically notified the administrator whenever the predefined threshold was exceeded.

These results demonstrate that centralized log management significantly improves an organization's ability to detect suspicious activities compared to manually reviewing system logs.

Although the planned integration with TheHive could not be completed because of hardware limitations, the project successfully demonstrated the complete detection phase of the incident response lifecycle. The proposed architecture shows how Graylog alerts can be forwarded to a case management platform to automate incident creation in larger production environments.

5.3 Achievement of Objectives

The objectives outlined at the beginning of the project were largely achieved.

Objective	Status
Design a virtual cybersecurity laboratory	Achieved
Configure Ubuntu Server	Achieved
Install and configure Graylog	Achieved
Configure centralized log collection	Achieved
Simulate SSH brute-force attacks	Achieved
Detect attacks using Event Definitions	Achieved
Generate automated security alerts	Achieved
Integrate Graylog with TheHive	Not Achieved

The only objective that was not fully implemented was the live integration with TheHive. However, the system architecture and workflow were successfully designed and documented.

5.4 Challenges Encountered

Several challenges were experienced during the implementation of this project.

Limited Hardware Resources

The most significant challenge was the limited hardware available for the virtual laboratory. The host computer had only **8 GB of RAM**, which was sufficient for running Graylog and the attack simulation but insufficient for running the full production deployment of TheHive.

Resource-Intensive Backend Services

The official TheHive deployment requires backend services such as Cassandra and Elasticsearch/OpenSearch. These services consume significant amounts of memory and repeatedly failed to start because the Linux operating system terminated them when available memory was exhausted.

Docker Configuration

Deploying Graylog and TheHive using Docker introduced additional challenges related to container networking, permissions, and service dependencies. These issues required several troubleshooting steps before Graylog could be successfully deployed.

Virtual Network Configuration

Configuring communication between the virtual machines also required adjustments to VMware networking. Initially, incorrect network adapter settings prevented communication between the systems. This was resolved by correctly configuring NAT and Host-only networking.

Event Definition Configuration

The Event Definition initially failed because the aggregation conditions were incomplete. After configuring the correct aggregation function, threshold, and time window, Graylog successfully detected brute-force attacks and generated alerts.

5.5 Lessons Learned

This project provided valuable practical experience in several areas of cybersecurity.

The implementation improved understanding of:

- Linux server administration
- Virtual machine management
- Network configuration
- SIEM deployment
- Centralized log management
- SSH authentication monitoring
- Attack simulation using Hydra
- Event correlation
- Security alerting
- Incident response workflows

The project also demonstrated that successful cybersecurity implementations require careful planning of both software architecture and hardware resources.

5.6 Recommendations

Based on the experience gained during this project, the following recommendations are made:

- Organisations should implement centralised log management instead of relying on manual log analysis.
- SIEM platforms should be configured with automated detection rules to reduce the time required to identify attacks.
- Incident response platforms such as TheHive should be integrated with SIEM systems to automate case creation and improve investigation efficiency.
- Future implementations should use servers with at least **16 GB of RAM** when deploying enterprise security platforms such as Graylog and TheHive together.
- Regular security monitoring should be combined with user awareness training and strong password policies to reduce the risk of successful brute-force attacks.

5.7 Future Work

Several improvements can be made to this project in future implementations.

These include:

- Full integration of Graylog with TheHive using HTTP Event Notifications.
- Deployment of Cortex for automated malware analysis and observable enrichment.
- Integration with threat intelligence platforms such as MISP.
- Monitoring additional log sources such as firewalls, Windows Event Logs, and web servers.
- Implementation of email notifications for high-priority security incidents.
- Development of custom Graylog dashboards for real-time security monitoring.
- Integration with Security Orchestration, Automation, and Response (SOAR) platforms.

These improvements would enhance the functionality of the proposed cybersecurity monitoring system and provide a more complete incident response solution.

5.8 Conclusion

Cybersecurity threats continue to increase in both frequency and complexity, making automated monitoring systems essential for protecting modern computer networks.

This project successfully demonstrated the implementation of a centralised log management and security monitoring solution using Graylog. By collecting authentication logs from an Ubuntu Server and

analysing them in real time, the system was able to detect SSH brute-force attacks and automatically generate security alerts.

The implementation showed that Graylog is an effective open-source SIEM platform for small and medium-sized organisations because it provides centralised visibility into security events and supports automated event detection.

Although the complete deployment of TheHive was not achieved due to hardware limitations, the project successfully designed the integration workflow and demonstrated how automated incident response can be achieved in a production environment.

Overall, the project met its primary objective of demonstrating how centralised log management, event detection, and automated alerting can improve cybersecurity monitoring and reduce the time required to identify potential attacks.

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APPENDICES

Appendix A

Hydra Command Used

```
hydra -l student -P passwords.txt ssh://192.168.101.137
```

Appendix B

SSH Status

```
sudo systemctl status ssh
```

Appendix C

Restart rsyslog

```
sudo systemctl restart rsyslog
```

Appendix D

Graylog Event Definition

Threshold

Count > 5

Time Window = 1 minute